

Untitled

October 4, 2026

```
[1]: import numpy as np
import pandas as pd
import kagglehub
```

```
[3]: path = kagglehub.dataset_download("yashdevladdha/uber-ride-analytics-dashboard")

print("Path to dataset files:", path)
```

Downloading to /home/annah/.cache/kagglehub/datasets/yashdevladdha/uber-ride-analytics-dashboard/2.archive...

100%|
| 16.5M/16.5M

[00:00<00:00, 27.7MB/s]

Extracting files...

Path to dataset files: /home/annah/.cache/kagglehub/datasets/yashdevladdha/uber-ride-analytics-dashboard/versions/2

```
[4]: import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
```

```
[6]: ls /home/annah/.cache/kagglehub/datasets/yashdevladdha/
↳ uber-ride-analytics-dashboard/versions/2
```

Dashboard.gif ncr_ride_bookings.csv Uber.pbix

```
[10]: df = pd.read_csv\
("/home/annah/.cache/kagglehub/datasets/yashdevladdha/
↳ uber-ride-analytics-dashboard/versions/2/ncr_ride_bookings.csv")
```

```
[11]: df.head()
```

```
[11]:
```

	Date	Time	Booking ID	Booking Status	Customer ID	\
0	2024-03-23	12:29:38	"CNR5884300"	No Driver Found	"CID1982111"	
1	2024-11-29	18:01:39	"CNR1326809"	Incomplete	"CID4604802"	
2	2024-08-23	08:56:10	"CNR8494506"	Completed	"CID9202816"	

3	2024-10-21	17:17:25	"CNR8906825"	Completed	"CID2610914"
4	2024-09-16	22:08:00	"CNR1950162"	Completed	"CID9933542"

	Vehicle Type	Pickup Location	Drop Location	Avg VTAT	Avg CTAT \
0	eBike	Palam Vihar	Jhilmil	NaN	NaN
1	Go Sedan	Shastri Nagar	Gurgaon Sector 56	4.9	14.0
2	Auto	Khandsa	Malviya Nagar	13.4	25.8
3	Premier Sedan	Central Secretariat	Inderlok	13.1	28.5
4	Bike	Ghitorni Village	Khan Market	5.3	19.6

	Reason for cancelling by Customer	Cancelled Rides by Driver \
0	...	NaN
1	...	NaN
2	...	NaN
3	...	NaN
4	...	NaN

	Driver Cancellation Reason	Incomplete Rides	Incomplete Rides Reason \
0	NaN	NaN	NaN
1	NaN	1.0	Vehicle Breakdown
2	NaN	NaN	NaN
3	NaN	NaN	NaN
4	NaN	NaN	NaN

	Booking Value	Ride Distance	Driver Ratings	Customer Rating \
0	NaN	NaN	NaN	NaN
1	237.0	5.73	NaN	NaN
2	627.0	13.58	4.9	4.9
3	416.0	34.02	4.6	5.0
4	737.0	48.21	4.1	4.3

	Payment Method
0	NaN
1	UPI
2	Debit Card
3	UPI
4	UPI

[5 rows x 21 columns]

```
[12]: df["driver_cancel"] = df["Booking Status"] == "Cancelled by Driver"
print(df.groupby("Pickup Location")["driver_cancel"].mean().describe())
print(df.groupby("Vehicle Type")["driver_cancel"].mean())
print(df.groupby(pd.to_datetime(df["Time"]).dt.hour)["driver_cancel"].mean())
```

count	176.000000
mean	0.180002
std	0.014011

```
min      0.145251
25%      0.170194
50%      0.179676
75%      0.188250
max      0.214689
```

Name: driver_cancel, dtype: float64

Vehicle Type

```
Auto      0.177530
Bike      0.181063
Go Mini    0.178823
Go Sedan   0.185365
Premier Sedan 0.179449
Uber XL    0.171274
eBike      0.180638
```

Name: driver_cancel, dtype: float64

```
/tmp/ipykernel_59219/3170267845.py:4: UserWarning: Could not infer format, so
each element will be parsed individually, falling back to `dateutil`. To ensure
parsing is consistent and as-expected, please specify a format.
```

```
print(df.groupby(pd.to_datetime(df["Time"]).dt.hour)["driver_cancel"].mean())
```

Time

```
0      0.180626
1      0.183824
2      0.165795
3      0.190166
4      0.181681
5      0.190955
6      0.168029
7      0.184587
8      0.177962
9      0.181200
10     0.179075
11     0.174136
12     0.178989
13     0.182998
14     0.186460
15     0.179468
16     0.179383
17     0.181365
18     0.182060
19     0.180864
20     0.180166
21     0.172158
22     0.185811
23     0.175959
```

Name: driver_cancel, dtype: float64

```
[15]: df["hour"] = pd.to_datetime(df["Time"], format="%H:%M:%S").dt.hour
df["dow"] = pd.to_datetime(df["Date"]).dt.dayofweek
```

```
[20]: print(df.groupby("Booking Status")["Avg VTAT"].apply(lambda s: s.isna().mean()))
print(df.groupby("driver_cancel")["Avg VTAT"].describe())
```

```
Booking Status
Cancelled by Customer    0.0
Cancelled by Driver      0.0
Completed               0.0
Incomplete              0.0
No Driver Found         1.0
Name: Avg VTAT, dtype: float64
```

	count	mean	std	min	25%	50%	75%	max
driver_cancel								
False	112500.0	8.685408	3.970272	2.0	5.4	8.5	11.9	20.0
True	27000.0	7.501952	2.600695	3.0	5.2	7.5	9.8	12.0

```
[24]: from sklearn.model_selection import StratifiedKFold
from sklearn.metrics import roc_auc_score
from catboost import CatBoostClassifier

y = df["driver_cancel"].astype(int)
cat_cols = ["Pickup Location", "Drop Location", "Vehicle Type"]
def cv_auc(features):
    X = df[features]
    cats = [c for c in features if c in cat_cols]
    aucs = []
    for tr, te in StratifiedKFold(5, shuffle=True, random_state=0).split(X, y):
        clf = CatBoostClassifier(cat_features=cats, verbose=0)
        clf.fit(X.iloc[tr], y.iloc[tr])
        aucs.append(roc_auc_score(y.iloc[te], clf.predict_proba(X.iloc[te]))[:,1]))
    return sum(aucs) / len(aucs)

feats_all = cat_cols + ["hour", "dow", "Avg VTAT"]
feats_no_vtat = cat_cols + ["hour", "dow"]
feats_vtat_only = ["Avg VTAT"]

print("all:      ", cv_auc(feats_all))
print("no VTAT:   ", cv_auc(feats_no_vtat))
print("VTAT only:  ", cv_auc(feats_vtat_only))
```

```
all:      0.6892983047274918
no VTAT:  0.5019511254140319
VTAT only: 0.6931202288467329
```

```
[ ]:
```